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Optimizing Error Messages: The Impact of Optimized Error Messages in Teaching F#

Bachelor's Thesis

by

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Declaration of Authorship

I hereby declare that the work I have submitted with the title „Optimizing Error Messages: The Impact of Optimized Error Messages in Teaching F#“ was independently written by me. I have fully disclosed all sources and aids used and have identified as quotations all passages of the work — including tables and figures — taken verbatim or in substance from other works or the internet, indicating the respective sources.

Kaiserslautern, on 8.2.2025

Jakob Meyerhöfer

Abstract

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Summary

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1. Introduction

1.1. Error Messages

1.1.1. Example 1 - `digitSum`

Task a) Implement a method `digitSum: Nat -> Nat`, that takes a natural number `n` as input and returns its digit sum. Examples:

```
digitSum 0N = 0N          digitSum 1234N = 10N
digitSum 123N = 6N        digitSum 4711N = 13N
```

```
let rec digitSum (n: Nat): Nat =
    if n = 0N then
        1N
    else
        (n % 10N) + digitSum (n / 10N)
```

Listing 1.1: *Student submission for `digitSum`*

```
let rec digitSumSolution (n: Nat): Nat =
    if n = 0N then
        0N
    else
        (n % 10N) + digitSumSolution (n / 10N)
```

Listing 1.2: *Sample solution for `digitSum`*

Example Tests - `digitSum`

```
    [<TestMethod>] [<Timeout(1000)>]
    member _.``a) digitSum Examples`` (): unit =
        test <@ digitSum 123N  = 6N @>
        test <@ digitSum 1234N = 10N @>
        test <@ digitSum 42N   = 6N @>
        test <@ digitSum 105N  = 6N @>
        test <@ digitSum 0N    = 0N @>
        test <@ digitSum 4711N = 13N @>
```

Listing 1.3: *Example tests for `digitSum`*

```
a) digitSum Examples (388ms): Error Message: Test method Tests+Tests.a
    ) digitSum Examples threw exception:
      Swensen.Unquote.AssertionFailedException: Test failed:

      StudentSubmission.digitSum 123N = 6N
      StudentSubmission.digitSum 123N = 6N
      7N = 6N
      false

      Stack Trace:
          at Tests.Tests.a) digitSum Examples() in path\to\Tests.fs:
              line 57
         at System.RuntimeMethodHandle.InvokeMethod(Object target,
             Void** arguments, Signature sig, Boolean isConstructor)
         at System.Reflection.MethodBaseInvoker.InvokeWithNoArgs(
             Object obj, BindingFlags invokeAttr)
```

Listing 1.4: *Output of examples tests for `digitSum`*

Key Information Example Tests - `digitSum`

Tested expression	`digitSum 123N`
Expected result	`6N`
Actual result	`7N`

Random Tests - `digitSum`

```
    [<TestMethod>] [<Timeout(5000)>]
    member _.``a) digitSum Random`` (): unit =
        Check.One (
            { Config.QuickThrowOnFailure with EndSize = 1000 },
            fun (n: Nat) -> Assert.AreEqual<Nat> (
                digitSumSolution n, digitSum n
            )
        )
```

Listing 1.5: *Random tests for `digitSum`*

```
a) digitSum Random (74ms): Error Message: Test method Tests+Tests.a) digitSum Random threw exception:
  System.Exception: Falsifiable, after 1 test (1 shrink) (StdGen (1765833463, 297438798)):
  Original:
  11N
  Shrunk:
  0N
  with exception:
  Microsoft.VisualStudio.TestTools.UnitTesting.AssertFailedException: Assert.AreEqual failed.
  Expected:<0>. Actual:<1>.
     at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.ThrowAssertFailed(String assertionName, String message) in /_/src/TestFramework/TestFramework/Assertions/Assert.cs:line 60
     at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.AreEqual[T](T expected, T actual, IEqualityComparer`1 comparer, String message, Object[] parameters) in /_/src/TestFramework/TestFramework/Assert
  ions/Assert.AreEqual.cs:line 201
     at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.AreEqual[T](T expected, T actual) in /_/src/TestFramework/TestFramework/Assertions/Assert.AreEqual.cs:line 34
     at Tests.a) digitSum Random@69.Invoke(Nat n) in path\to\Tests.fs:line 69
     at FsCheck.Testable.evaluate[a,b](FSharpFunc`2 body, a a)

  Stack Trace:
      at Microsoft.FSharp.Core.PrintfModule.PrintFormatToStringThenFail@1448.Invoke(String message)
     at <StartupCode$FsCheck>.$Runner.get_throwingRunner@382.FsCheck.IRunner.OnFinished(String name, TestResult testResult)
     at FsCheck.Runner.runner[a](Config config, a prop)
     at FsCheck.Runner.check[a](Config config, a p)
     at FsCheck.Check.One[Testable](Config config, Testable property)
     at Tests.Tests.a) digitSum Random() in path\to\Tests.fs:line 68
     at System.RuntimeMethodHandle.InvokeMethod(Object target, Void** arguments, Signature sig, Boolean isConstructor)
     at System.Reflection.MethodBaseInvoker.InvokeWithNoArgs(Object obj, BindingFlags invokeAttr)
```

Listing 1.6: *Output of random tests for `digitSum`*

Key Information Random Tests - `digitSum`

Tested expression	`a) digitSum Random Original: 11N` } `digitSum 11N`
Expected result	`Expected:<0>` $\Longrightarrow$ `0` (actually `0N`)
Actual result	`Actual:<1>` $\Longrightarrow$ `1` (actually `1N`)

1.1.2. Example 2 - `sortedDigits`

Task b) Implement a method `sortedDigits: Nat -> bool`, that takes a natural number `n` as input and returns `true` if the digits are in ascending order and `false` otherwise. Examples:

```
sortedDigits 0N = true          sortedDigits 1111N = true
sortedDigits 5N = true          sortedDigits 1101N = false
sortedDigits 159N = true        sortedDigits 543N = false
```

```
let rec sortedDigits (n: Nat): bool =
    if n = 0N then
        true
    else
        let frontSorted: bool = sortedDigits (n / 10N)
        let lastDigit: Nat = n % 10N
        let secondLastDigit: Nat = (n / 10N) % 10N
        frontSorted && secondLastDigit < lastDigit
```

Listing 1.7: *Student submission for `sortedDigits`*

```
let rec sortedDigitsSolution (n: Nat): bool =
    if n = 0N then
        true
    else
        let frontSorted: bool = sortedDigitsSolution (n / 10N)
        let lastDigit: Nat = n % 10N
        let secondLastDigit: Nat = (n / 10N) % 10N
        frontSorted && secondLastDigit ≤ lastDigit
```

Listing 1.8: *Sample solution for `sortedDigits`*

Example Tests - `sortedDigits`

```
    [<TestMethod>] [<Timeout(1000)>]
    member _.``b) sortedDigits Examples`` (): unit =
        test <@ sortedDigits 0N    = true @>
        test <@ sortedDigits 5N    = true @>
        test <@ sortedDigits 159N  = true @>
        test <@ sortedDigits 1111N = true @>
        test <@ sortedDigits 42N   = false @>
        test <@ sortedDigits 543N  = false @>
        test <@ sortedDigits 1101N = false @>
```

Listing 1.9: *Example tests for `sortedDigits`*

```
b) sortedDigits Examples (4ms): Error Message: Test method Tests+Tests
    .b) sortedDigits Examples threw exception:
      Swensen.Unquote.AssertionFailedException: Test failed:

      StudentSubmission.sortedDigits 1111N = true
      StudentSubmission.sortedDigits 1111N = true
      false = true
      false

      Stack Trace:
          at Tests.Tests.b) sortedDigits Examples() in path\to\Tests.
              fs:line 80
         at System.RuntimeMethodHandle.InvokeMethod(Object target,
             Void** arguments, Signature sig, Boolean isConstructor)
         at System.Reflection.MethodBaseInvoker.InvokeWithNoArgs(
             Object obj, BindingFlags invokeAttr)
```

Listing 1.10: *Output of examples tests for `sortedDigits`*

Key Information Example Tests - `sortedDigits`

Tested expression	`sortedDigits 123N`
Expected result	`true`
Actual result	`false`

Random Tests - `sortedDigits`

```
    [<TestMethod>] [<Timeout(5000)>]
    member _.``b) sortedDigits Random`` (): unit =
        Check.One (
            { Config.QuickThrowOnFailure with EndSize = 1000 },
            fun (n: Nat) -> Assert.AreEqual<bool> (
                sortedDigitsSolution n, sortedDigits n
            )
        )
```

Listing 1.11: *Random tests for `sortedDigits`*

```
b) sortedDigits Random (3ms): Error Message: Test method Tests+Tests.b)
    sortedDigits Random threw exception:
      System.Exception: Falsifiable, after 5 tests (0 shrinks) (StdGen
           (867569077, 297439050)):
      Original:
      33N
      with exception:
      Microsoft.VisualStudio.TestTools.UnitTesting.
          AssertFailedException: Assert.AreEqual failed.
      Expected:<True>. Actual:<False>.
         at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.
             ThrowAssertFailed(String assertionName, String message)
             in /_/src/TestFramework/TestFramework/Assertions/Assert.
             cs:line 60
         at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.
             AreEqual[T](T expected, T actual, IEqualityComparer`1
             comparer, String message, Object[] parameters) in /_/src/
             TestFramework/TestFramework/Assert
      ions/Assert.AreEqual.cs:line 201
         at Microsoft.VisualStudio.TestTools.UnitTesting.Assert.
             AreEqual[T](T expected, T actual) in /_/src/TestFramework
             /TestFramework/Assertions/Assert.AreEqual.cs:line 34
         at Tests.b) sortedDigits Random@90.Invoke(Nat n) in path\to\
             Tests.fs:line 90
         at FsCheck.Testable.evaluate[a,b](FSharpFunc`2 body, a a)

      Stack Trace:
          at Microsoft.FSharp.Core.PrintfModule.
              PrintFormatToStringThenFail@1448.Invoke(String message)
         at <StartupCode$FsCheck>.$Runner.get_throwingRunner@382.
             FsCheck.IRunner.OnFinished(String name, TestResult
             testResult)
         at FsCheck.Runner.runner[a](Config config, a prop)
         at FsCheck.Runner.check[a](Config config, a p)
         at FsCheck.Check.One[Testable](Config config, Testable
             property)
         at Tests.Tests.b) sortedDigits Random() in path\to\Tests.fs:
             line 89
         at System.RuntimeMethodHandle.InvokeMethod(Object target,
             Void** arguments, Signature sig, Boolean isConstructor)
         at System.Reflection.MethodBaseInvoker.InvokeWithNoArgs(
             Object obj, BindingFlags invokeAttr)
```

Listing 1.12: *Output of random tests for `sortedDigits`*

Key Information Random Tests - `sortedDigits`

Tested expression	`b) sortedDigits Random Original: 33N` } `sortedDigits 33N`
Expected result	`Expected:<True>` $\Longrightarrow$ `True` (actually `true`)
Actual result	`Actual:<False>` $\Longrightarrow$ `False` (actually `false`)

2. Related Work

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3.1. Else

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4. Conclusion

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List of Figures

A. My Code

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B. My Ideas

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